

REMARKS ON SIX-TANGENTIAL CONICS TO SMOOTH PLANE SEXTICS

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ABSTRACT.

We work over the complex number field $\mathbb{C}$.

Let B be a smooth plane sextic. A smooth conic C is called a six-tangential conic to B if C is tangent to B at six distinct points. This paper is motivated by the following simple observation:

Proposition 0.1. *For $\nu = 0, 2, 4, 6$, let $f_\nu(x_0, x_1)$ be a general homogeneous polynomial in variables x_0, x_1 of degree ν . Then the smooth plane sextic B defined by the equation*

$$(0.1) \quad f_6(x_0, x_1) + f_4(x_0, x_1)x_2^2 + f_2(x_0, x_1)x_2^4 + f_0(x_0, x_1)x_2^6 = 0$$

possesses exactly 120 six-tangential conics.

See Section ?? for the proof of this proposition and further geometric properties of this configuration of 120 conics.

Let B be a smooth plane sextic. A smooth conic C is called a splitting conic to B if the local intersection multiplicity of B and C at every point of $B \cap C$ is even. Let $\pi_B: X_B \rightarrow \mathbb{P}^2$ be the double cover branched along B . Then X_B is a K3 surface. A smooth rational curve on X_B is called a (-2) -curve. Let C be a splitting conic of B . Then the inverse image $\pi_B^{-1}(C)$ is the union of two smooth rational curves $\tilde{C}^+$ and $\tilde{C}^-$. Let C_1 and C_2 be two distinct splitting conics of B . Then we have

$$\langle \tilde{C}_1^+, \tilde{C}_2^+ \rangle \in \{0, 1, 2, 3, 4\}, \quad \langle \tilde{C}_1^+, \tilde{C}_2^+ \rangle + \langle \tilde{C}_1^+, \tilde{C}_2^- \rangle = 4.$$

Let k be an integer with $k \geq 2$. Let E_k be the set of two-element subsets of $\{1, \dots, k\}$, and let W_k denote the set of mappings

$$w: E_k \rightarrow \{0, 1, 2, 3, 4\}.$$

Thus W_k is the set of weighted complete graphs on k vertices $\{1, \dots, k\}$ with weights $0, 1, 2, 3, 4$. Then the symmetric group $\mathfrak{S}_k$ acts on W_k via its natural action on the vertex set $\{1, \dots, k\}$:

$${}^\tau w(\{i, j\}) := w(\{i^\tau, j^\tau\}).$$

We let the multiplicative group $\{\pm 1\}$ act on $\{0, 1, 2, 3, 4\}$ by

$$a^s := \begin{cases} a & \text{if } s = 1, \\ 4 - a & \text{if } s = -1. \end{cases}$$

Then $\{\pm 1\}^k$ acts on W_k by

$$(s_1, \dots, s_k)w(\{i, j\}) = (w(\{i, j\}))^{s_i s_j}.$$

Thus the semidirect product $G := \{\pm 1\}^k \rtimes \mathfrak{S}_k$ acts on W_k . We call a G -orbit in W_k a splitting graph.

Remark 0.2. The group $\mathfrak{S}_k$ acts on $\{\pm 1\}^k$ by

$$({}^\tau s)_i := s_{i^\tau}.$$

The product in G is given by

$$(s, \tau)(s', \tau') = (s {}^\tau s', \tau \tau').$$

The action of G on W_k is given by

$$(s, \tau)w = s({}^\tau w).$$

Remark 0.3. Our construction is a variant of Shirane's splitting graph adapted to collections of splitting conics.

Suppose that a smooth plane sextic B has a collection Γ of k distinct splitting conics. By labelling these splitting conics as $C_1, \dots, C_k$, and choosing an irreducible component $\tilde{C}_i^+$ from the two irreducible components of $\pi_B^{-1}(C_i)$ for each $i = 1, \dots, k$, we obtain an element w_Γ of W_k by

$$w_\Gamma(\{i, j\}) := \langle \tilde{C}_i^+, \tilde{C}_j^+ \rangle.$$

Note that the G -orbit $[w_\Gamma] := G \cdot w_\Gamma$ is independent of the labelling and of the choices of the irreducible components $\tilde{C}_i^+$.

Theorem 0.4. *Let γ be a splitting graph in the first column of Table ?? . Then the family $\mathcal{F}_\gamma \subset \mathbb{P}_*(\Gamma(\mathbb{P}^2, \mathcal{O}(6)))$ of smooth plane sextics B possessing a collection Γ of splitting conics such that $[w_\Gamma] = \gamma$ is irreducible, and a general member of $\mathcal{F}_\gamma$ has a collection $\Gamma' \supset \Gamma$ of splitting conics such that $[w_{\Gamma'}] = \gamma'$, where γ' is the splitting graph in the same row of Table ?? as γ .*

For example, the first row of Table ?? has the following interpretation. Let B be a general plane sextic in the family of smooth plane sextics possessing two splitting conics C_1 and C_2 such that $\langle \tilde{C}_1^+, \tilde{C}_2^+ \rangle = 0$ for a suitable choice of the lifts $\tilde{C}_1^+, \tilde{C}_2^+$. Then B has another splitting conic C_3 such that

$$\langle \tilde{C}_1^+, \tilde{C}_3^+ \rangle = \langle \tilde{C}_2^+, \tilde{C}_3^+ \rangle = 0$$

for a suitable choice of $\tilde{C}_3^+$.

0.1. Enumeration of splitting conics. Let B be a smooth plane sextic, let $\pi_B: X_B \rightarrow \mathbb{P}^2$ be the double cover branched along B , and let $j: X_B \rightarrow X_B$ be the involution with respect to the double cover $\pi_B: X_B \rightarrow \mathbb{P}^2$. We denote by $h \in \text{Pic}(X_B)$ the class of the pull-back of a line of $\mathbb{P}^2$. Then h is ample. The involution j fixes h and acts as multiplication by -1 on the orthogonal complement $h^\perp$ in $\text{Pic}(X_B)$. The set $\mathcal{R}_d$ of classes of smooth rational curves C of degree $\langle C, h \rangle = d$ on X_B is enumerated inductively on d as follows:

$$\mathcal{R}_d := \left\{ r \in \text{Pic}(X_B) \mid \begin{array}{l} \langle r, r \rangle = -2, \langle r, h \rangle = d, \\ \text{for all } d' < d \text{ and for all } r' \in \mathcal{R}_{d'}, \text{ we have } \langle r, r' \rangle \geq 0 \end{array} \right\}.$$

Then the set of classes of splitting conics of B is equal to the set of j -orbits in $\mathcal{R}_2$.

0.2. Proof of Proposition 0.1. Throughout this subsection, let B denote a smooth plane sextic defined by an equation of the form (0.1). Let $\iota: \mathbb{P}^2 \rightarrow \mathbb{P}^2$ be the involution given by

$$[x_0 : x_1 : x_2] \mapsto [x_0 : x_1 : -x_2].$$

Then B is ι -invariant. We show that the Picard lattice $\text{Pic}(X_B)$ of the $K3$ surface X_B is of rank 9 and admits a basis such that

(a) the Gram matrix of $\text{Pic}(X_B)$ is the diagonal matrix

$$\text{diag}(2, -2, \dots, -2),$$

(b) the class $h \in \text{Pic}(X_B)$ of the pull-back of a line of $\mathbb{P}^2$ is written as

$$h = (3, -1, \dots, -1).$$

We denote the space of all plane sextics by

$$\mathcal{P}_6 := \mathbb{P}_*(\Gamma(\mathbb{P}^2, \mathcal{O}(6))),$$

and let $\mathcal{S} \subset \mathcal{P}_6$ be the Zariski open subset of smooth plane sextics. Then the period map gives an open immersion of $\mathcal{S}/\text{PGL}(3, \mathbb{C})$ into a 19-dimensional period domain. The space $\mathcal{I}_\iota$ of ι -invariant sextics is a 15-dimensional subspace of $\mathcal{P}_6$. Since the centralizer of ι in $\text{PGL}(3, \mathbb{C})$ is of dimension 4, the image of

$$\mathcal{I}_\iota \cap \mathcal{S} \hookrightarrow \mathcal{S} \rightarrow \mathcal{S}/\text{PGL}(3, \mathbb{C})$$

is $15 - 4 = 11$. (Here we use the fact that a smooth plane sextic is of genus 10 and hence its automorphism group is finite.) Since B is a general point of $\mathcal{I}_\iota \cap \mathcal{S}$, the rank of $\text{Pic}(X_B)$ is $20 - 11 = 9$. Let $\phi: \mathbb{P}^2 \rightarrow Q$ be the quotient morphism by the involution ι , where $Q \subset \mathbb{P}^3$ is the quadric surface of rank 3 defined by $z_0 z_2 - z_1^2 = 0$, and ϕ is given by

$$[x_0 : x_1 : x_2] \mapsto [z_0 : z_1 : z_2 : z_3] = [x_0^2 : x_0 x_1 : x_1^2 : x_2^2].$$

The fixed point $[0 : 0 : 1]$ of ι is mapped to the vertex A of Q and the fixed line $\{x_2 = 0\}$ is mapped to the plane section $Q \cap H_0$ of Q , where $H_0 = \{z_3 = 0\}$.

The quotient morphism ϕ induces a double covering $B \rightarrow B_Q$ of a $(2, 3)$ -complete intersection $B_Q \subset Q$. Let $\varpi_B: Y_B \rightarrow Q$ be the double covering branched along B_Q and at the vertex A . Then Y_B is a del Pezzo surface of degree 1, and the morphism $\varpi_B: Y_B \rightarrow Q$ is the bi-anticanonical model of Y_B . We denote by α_W the anti-canonical class of Y_B . The morphism ϕ lifts to a double covering $\varphi: X_B \rightarrow Y_B$. Thus we obtain the following commutative diagram:

$$\begin{array}{ccc} X_B & \xrightarrow{\varphi} & Y_B \\ \pi_B \downarrow & & \downarrow \varpi_B \\ \mathbb{P}^2 & \xrightarrow[\phi]{} & Q. \end{array}$$

The branch locus of the double covering $\varphi: X_B \rightarrow Y_B$ is the pull-back $\varpi_B^{-1}(Q \cap H_0)$, which is a member of the bi-anticanonical system $|2\alpha_W|$. Since the pull-back by ϕ of a plane section $Q \cap H$ of Q is a conic on $\mathbb{P}^2$, we have $2h = \varphi^*(2\alpha_W)$. Since $\text{Pic}(X_B)$ is torsion free, we obtain

$$h = \varphi^*\alpha_W.$$

The del Pezzo surface Y_B is obtained by blowing up a projective plane $\mathbf{P}^2$ at eight distinct points in general position. Hence $\text{Pic}(Y_B)$ is isomorphic to $I_{1,8}$, and has a basis $l, e_1, \dots, e_8$, where l is the class of the pull-back of a line of $\mathbf{P}^2$ and $e_1, \dots, e_8$ are the classes of exceptional divisors. We have

$$\alpha_W = 3l - e_1 - \dots - e_8.$$

Now we have

$$I_{1,8}(2) \cong \varphi^*\text{Pic}(Y_B) \subset \text{Pic}(X_B).$$

Since $\text{Pic}(X_B)$ is of rank 9, $\text{Pic}(X_B)$ is an overlattice of $I_{1,8}(2)$. The discriminant form of $I_{1,8}(2)$ has 135 non-zero isotropic vectors. By Nikulin's theory of discriminant forms [?], we see that $I_{1,8}(2)$ has $1 + 135 + 1575 + 2025 + 270$ even overlattices as in Table 0.1. Note that

$$h = \varphi^*(3l - e_1 - \dots - e_8)$$

is ample, and hence

$$R(h^\perp) := \{r \in \text{Pic}(X_B) \mid \langle r, r \rangle = -2, \langle r, h \rangle = 0\}$$

should be empty. On the other hand, as is shown in Table 0.1, if the extension index $\text{Pic}(X_B)/I_{1,8}(2)$ is > 1 , then $R(h^\perp)$ would have an element. Therefore the overlattice $\text{Pic}(X_B)$ is in fact equal to $\varphi^*\text{Pic}(Y_B)$ and hence has a basis $\varphi^*l, \varphi^*e_1, \dots, \varphi^*e_8$ with the desired properties (a), (b).

Then we see that X_B has exactly 120 splitting conics by the method given in Section 0.1. Indeed, there are no vectors r in $\text{Pic}(X_B)$ with $\langle r, r \rangle = -2$ and $\langle h, r \rangle = 1$, and there exists exactly 240 vectors r with $\langle r, r \rangle = -2$ and $\langle h, r \rangle = 2$. Therefore these 240 vectors are the classes of smooth rational curves. Moreover

$\dim_{\mathbb{F}_2} M/I_{1,8}(2)$	number	type of $R(h^\perp)$
0	1	0
1	135	$8A_1$
2	1575	$2D_4$
3	2025	D_8
4	270	E_8

TABLE 0.1. Overlattices M of $I_{1,8}(2)$

the involution j of the double covering $X_B \rightarrow \mathbb{P}^2$ acts on these 240 vectors without fixed points, and hence the 120 orbits under the involution j are in one-to-one correspondence with splitting conics.

In fact, these 240 vectors r are the pull-back of the 240 lines of the del Pezzo surface Y_B . Note that the mapping $B \mapsto B_Q$ gives an isomorphism between the space the projective space $\mathcal{I}_\iota$ of ι -invariant sextics and the linear system $|\mathcal{O}_Q(3)|$. Hence Y_B is a general member of the family $\mathcal{U}$ of bi-anticanonical models of del Pezzo surfaces of degree 1 studied in [?]. By [?, Proposition 4.4], the union of the 240 lines of Y_B has only ordinary double points, which implies that the 120 splitting conics of B are in fact six-tangential conics.

Remark 0.5. In fact, the automorphism group of the $K3$ surface X_B is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and the 240 curves obtained above are all the smooth rational curves on X_B .

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